



(b)

From/to	Midfield	Box	Wing	Goal	Lost
Midfield	25%	20%	10%	5%	40%
Box	10%	25%	20%	15%	30%
Wing	10%	10%	25%	5%	50%
Goal	0%	0%	0%	100%	0%
Lost	0%	0%	0%	0%	100%

transition matrix

$$\begin{pmatrix} x_{mid} \\ x_{wing} \\ x_{box} \end{pmatrix} = \begin{pmatrix} 25\% & 10\% & 20\% \\ 10\% & 25\% & 20\% \\ 10\% & 10\% & 25\% \end{pmatrix} \begin{pmatrix} x_{mid} \\ x_{wing} \\ x_{box} \end{pmatrix} + \begin{pmatrix} 5\% \\ 5\% \\ 15\% \end{pmatrix}$$

3×1 3×3 (A) 3×1 (X) 3×1 (g)

Exam: Transition matrix & eq forming

$$X = AX + g$$

$$\Rightarrow X - AX = g$$

$$\Rightarrow (I - A)X = g$$

$$\boxed{X}$$

x_{mid} = starting at mid, probability of scoring

$$\Rightarrow x_{mid} = 5\% + 25\%x_{mid} + 10\%x_{wing} + 20\%x_{box}$$

$$x_{wing} = 5\% + 10\%x_{mid} + 25\%x_{wing} + 10\%x_{box}$$

$$x_{box} = 15\% + 10\%x_{mid} + 20\%x_{wing} + 25\%x_{box}$$